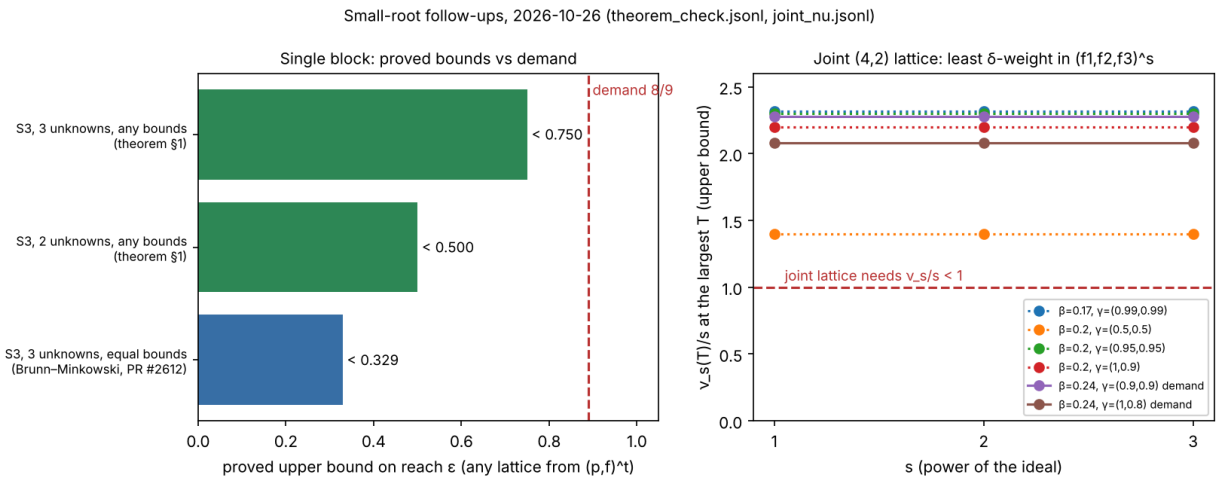
Small-root follow-ups

Small-root decomposition, follow-ups: unequal bounds, non-convex monomial sets, the joint (4,2) lattice

2026-10-26 · follow-up to `../smallroot_optimal_reach_20261026/` (PR #2612) · tools `examples/smallroot_optimal_reach.rs` (`theorem` mode) and `examples/smallroot_joint_lattice.rs` (std only) · records `theorem_check.jsonl`, `joint_nu.jsonl`, `joint_control.jsonl` · graph `followups.svg` · PDF `report.pdf`

Class: analysis (a proof, an exact check of it, and a computation of an ideal invariant). Not an index-calculus measurement, no leaderboard row.



Verdict

open item from PR #2612	result	status
unequal per-variable bounds	any lattice from `(p,f)^t` needs `h_N(δ) = max_{v∈supp f} ⟨δ,v⟩ < 1`: **ε < 3/4** for S_3 in 3 unknowns, **ε < 1/2** in 2 unknowns, against a demand of 8/9	**proved** (§1); exact check 9,600 cases, 0 violations (§2)
monomial sets that are not convex	the same theorem holds for **every** monomial set `M`	**proved** (§1), same check on random non-convex `M`
one joint lattice for the three-equation `(4,2)` system	the joint lattice needs `ν_s < s` for some `s` (`ν_s` = least δ-weight in $(\bar f_1, \bar f_2, \bar f_3)^s$); computed `ν_s/s = min_k deg_δ f_k` ≥ **2.08** in the demand region for `s ≤ 3`	**computed, not proved** (§3): upper bounds from truncated Macaulay matrices, stable in the truncation; positive control passes

With PR #2612 this makes the windowed-decomposition (small-root) family closed for every single-block Coppersmith lattice, for any bound shape and any monomial set, by a proof; the joint three-block lattice is closed by computation for `s ≤ 3` with a factor-two margin, not by proof.

1. Theorem (any monomial set, any bounds)

Let `f ∈ Z[x₁..x_n]` with $\bar f$ `= f mod p ≠ 0`, Newton support `N`, and bounds `|x_i| < p^{δ_i}`, `δ ≥ 0`. Let `L` be any lattice of polynomials in `(p, f)^t` supported on a finite monomial set `M` (Jochemsz–May, Ryan's optimal shifts, any strategy; `M` arbitrary), and `E = log_p [L : p^t Z^M]`. If the Howgrave-Graham determinant condition `Σ_{μ∈M} ⟨δ, μ⟩ < E` holds, then

> `h_N(δ) := max_{v∈N} ⟨δ, v⟩ < 1.`

Proof. (i) Filter `L` by `L_i = L ∩ p^i Z^M`. `p, f` is a regular sequence, so `((p,f)^t : p^i) = (p,f)^{t−i}` and `L_i/L_{i+1}` embeds, by `g ↦ g/p^i mod p`, into `W_{t−i} = f̄^{t−i} F_p[x] ∩ span M`. Hence `E ≤ Σ_{s=1}^{t} dim W_s`. (ii) Fix a monomial order refining the weight `δ` (ties broken by any monomial order). An echelon basis of `W_s` has distinct leading monomials; each is the leading monomial of some `h·f̄^s`, whose δ-weight is `deg_δ(h f̄^s) = deg_δ h + s·deg_δ f̄ ≥ s·h_N(δ)` (`F_p[x]` is a domain, so δ-initial forms multiply). So a monomial `μ` is a leading monomial in at most `⌊⟨δ,μ⟩ / h_N(δ)⌋` of the layers, and `E ≤ Σ_{μ∈M} ⟨δ,μ⟩ / h_N(δ)`. (iii) With the condition, `Σ⟨δ,μ⟩ < E ≤ Σ⟨δ,μ⟩ / h_N(δ)`, so `h_N(δ) < 1`. ∎

No convexity is used. For S_3 in three unknowns the vertices `(2,2,0)`, `(2,0,2)`, `(0,2,2)` give `2(δ_i + δ_j) < 1` for all pairs, so `ε = Σδ_i < 3/4`; in two unknowns the vertex `(2,2)` gives `ε < 1/2`, which is the value the PR #2612 optimiser approached (0.489). The bound is sharp in the univariate case (`dδ < 1`, Coppersmith's `1/d`). For three unknowns and equal bounds the Brunn–Minkowski bound of PR #2612 (0.329, convex `M`) is stronger.

Scope: the determinant (Howgrave-Graham) criterion that every Coppersmith strategy, Ryan's included, is judged by; dimension factors are dropped, which favours the attack. It is not a statement about lattices that happen to contain unusually short vectors beyond the determinant estimate.

2. Exact check of the theorem's key inequality

`smallroot_optimal_reach theorem` draws random monomial sets (each monomial of a box kept with probability 0.45–0.85, so almost never convex), computes `E(M)` exactly by elimination over `Z/p^t` (`p = 1000003`, superset box four wider), and checks `E(M) ≤ Σ_μ min(t, ⌊⟨w,μ⟩/h_N(w)⌋)` for 12 random weights each.

f	unknowns	box	random sets	with E > 0	weight checks	violations	max E / bound	max λ*·h_N
linear	2	6	200	200	2400	**0**	0.8438	0.4599
univariate-d2	1	8	200	190	2400	**0**	1.0000	0.8571
s3	2	6	200	170	2400	**0**	0.6897	0.3896
s3	3	4	200	68	2400	**0**	0.2373	0.1181

`λ*·h_N(w) ≤ 1` is the theorem's conclusion in scaled form (`λ* = E/Σ⟨w,μ⟩`); `max E/bound = 1.0000` for the univariate case shows the inequality is attained.

3. The joint lattice for the three-equation (4,2) system

$f_1 = S_3(x_1,x_2,u)$, $f_2 = S_3(x_3,x_4,v)$, $f_3 = S_3(u,v,x_R)$; bounds p^{β} on the leaves, p^{γ_u}, p^{γ_v} on u, v. The demand (CLOSURE.md §1) is $\beta < 1/4$ and $3\beta + \gamma_u + \gamma_v > 5/2$. Steps (i)–(ii) of §1 carry over to the ideal $(p, f_1, f_2, f_3)^t$ (assuming $\tilde f_1, \tilde f_2, \tilde f_3$ is a regular sequence, which the tree structure suggests), with $W_s = \bar I^s \cap \mathrm{span}\, M$: a monomial counted in layer s has δ-weight at least $\nu_s = \min\{\deg_\delta g : 0 \neq g \in \bar I^s\}$. So the joint lattice can meet the condition only if **$\nu_s < s$ for some s**.

For one block $\nu_s = s\cdot h_N(\delta)$. Here the δ-initial forms of f_1 and f_3 share u^2 ($u^2(x_1-x_2)^2$ and u^2v^2), and those of f_2 and f_3 share v^2, so the single-block argument does not apply: combinations such as $v^2 f_1 - (x_1-x_2)^2 f_3$ lose their top weight. No weight $w \le \delta$ with regular initial forms has $\deg_w f_k \ge 1$ for all k (the leaves force $4\beta < 1$), so no simple weight argument closes it.
`smallroot_joint_lattice` computes $\nu_s(T)$, the least δ-weight in the row space of the Macaulay matrix $\{x^a f^\alpha : |\alpha| = s, \deg \le T\}$ over $\mathbb{F}_p$. Truncation only drops ideal elements, so $\nu_s(T)$ is an **upper bound** on ν_s.

β	γ_u	γ_v	in demand region	deg_δ f1, f2, f3	s	T	rows × cols	ν_s(T)	ν_s/s
0.17	0.99	0.99	no	2.32, 2.32, 3.96	1	6	84 × 924	2.32	2.32
0.17	0.99	0.99	no	2.32, 2.32, 3.96	1	8	630 × 3003	2.32	2.32
0.17	0.99	0.99	no	2.32, 2.32, 3.96	1	10	2772 × 8008	2.32	2.32
0.17	0.99	0.99	no	2.32, 2.32, 3.96	2	8	6 × 3003	4.64	2.32
0.17	0.99	0.99	no	2.32, 2.32, 3.96	2	10	168 × 8008	4.64	2.32
0.17	0.99	0.99	no	2.32, 2.32, 3.96	2	12	1260 × 18564	4.64	2.32
0.17	0.99	0.99	no	2.32, 2.32, 3.96	3	12	10 × 18564	6.96	2.32
0.17	0.99	0.99	no	2.32, 2.32, 3.96	3	13	70 × 27132	6.96	2.32
0.2	0.95	0.95	no	2.30, 2.30, 3.80	1	6	84 × 924	2.30	2.30
0.2	0.95	0.95	no	2.30, 2.30, 3.80	1	8	630 × 3003	2.30	2.30
0.2	0.95	0.95	no	2.30, 2.30, 3.80	1	10	2772 × 8008	2.30	2.30
0.2	0.95	0.95	no	2.30, 2.30, 3.80	2	8	6 × 3003	4.60	2.30
0.2	0.95	0.95	no	2.30, 2.30, 3.80	2	10	168 × 8008	4.60	2.30
0.2	0.95	0.95	no	2.30, 2.30, 3.80	2	12	1260 × 18564	4.60	2.30
0.2	0.95	0.95	no	2.30, 2.30, 3.80	3	12	10 × 18564	6.90	2.30
0.2	0.95	0.95	no	2.30, 2.30, 3.80	3	13	70 × 27132	6.90	2.30
0.24	0.9	0.9	yes	2.28, 2.28, 3.60	1	6	84 × 924	2.28	2.28
0.24	0.9	0.9	yes	2.28, 2.28, 3.60	1	8	630 × 3003	2.28	2.28
0.24	0.9	0.9	yes	2.28, 2.28, 3.60	1	10	2772 × 8008	2.28	2.28
0.24	0.9	0.9	yes	2.28, 2.28, 3.60	2	8	6 × 3003	4.56	2.28
0.24	0.9	0.9	yes	2.28, 2.28, 3.60	2	10	168 × 8008	4.56	2.28
0.24	0.9	0.9	yes	2.28, 2.28, 3.60	2	12	1260 × 18564	4.56	2.28
0.24	0.9	0.9	yes	2.28, 2.28, 3.60	3	12	10 × 18564	6.84	2.28
0.24	0.9	0.9	yes	2.28, 2.28, 3.60	3	13	70 × 27132	6.84	2.28

β	γ_u	γ_v	in demand region	deg_δ f_1, f_2, f_3	s	T	rows × cols	ν_s(T)	ν_s/s
0.2	1	0.9	no	2.40, 2.20, 3.80	1	6	84 × 924	2.20	2.20
0.2	1	0.9	no	2.40, 2.20, 3.80	1	8	630 × 3003	2.20	2.20
0.2	1	0.9	no	2.40, 2.20, 3.80	1	10	2772 × 8008	2.20	2.20
0.2	1	0.9	no	2.40, 2.20, 3.80	2	8	6 × 3003	4.40	2.20
0.2	1	0.9	no	2.40, 2.20, 3.80	2	10	168 × 8008	4.40	2.20
0.2	1	0.9	no	2.40, 2.20, 3.80	2	12	1260 × 18564	4.40	2.20
0.2	1	0.9	no	2.40, 2.20, 3.80	3	12	10 × 18564	6.60	2.20
0.2	1	0.9	no	2.40, 2.20, 3.80	3	13	70 × 27132	6.60	2.20
0.24	1	0.8	yes	2.48, 2.08, 3.60	1	6	84 × 924	2.08	2.08
0.24	1	0.8	yes	2.48, 2.08, 3.60	1	8	630 × 3003	2.08	2.08
0.24	1	0.8	yes	2.48, 2.08, 3.60	1	10	2772 × 8008	2.08	2.08
0.24	1	0.8	yes	2.48, 2.08, 3.60	2	8	6 × 3003	4.16	2.08
0.24	1	0.8	yes	2.48, 2.08, 3.60	2	10	168 × 8008	4.16	2.08
0.24	1	0.8	yes	2.48, 2.08, 3.60	2	12	1260 × 18564	4.16	2.08
0.24	1	0.8	yes	2.48, 2.08, 3.60	3	12	10 × 18564	6.24	2.08
0.24	1	0.8	yes	2.48, 2.08, 3.60	3	13	70 × 27132	6.24	2.08
0.2	0.5	0.5	no	1.40, 1.40, 2.00	1	6	84 × 924	1.40	1.40
0.2	0.5	0.5	no	1.40, 1.40, 2.00	1	8	630 × 3003	1.40	1.40
0.2	0.5	0.5	no	1.40, 1.40, 2.00	1	10	2772 × 8008	1.40	1.40
0.2	0.5	0.5	no	1.40, 1.40, 2.00	2	8	6 × 3003	2.80	1.40
0.2	0.5	0.5	no	1.40, 1.40, 2.00	2	10	168 × 8008	2.80	1.40
0.2	0.5	0.5	no	1.40, 1.40, 2.00	2	12	1260 × 18564	2.80	1.40
0.2	0.5	0.5	no	1.40, 1.40, 2.00	3	12	10 × 18564	4.20	1.40
0.2	0.5	0.5	no	1.40, 1.40, 2.00	3	13	70 × 27132	4.20	1.40

Positive control (`--control`): with generators $\{f_1, f_2, f_1 + x_1\}$ the ideal contains x_1, so ν_1 must equal $\beta = 0.2$; the code returns 0.2000 at $T = 6$ and $T = 8$ (`joint_control.jsonl`). Cancellation is visible to the instrument.

Reading. In every cell $\nu_s(T) = s \cdot \min_k \deg_\delta f_k$ exactly, for $s = 1, 2, 3$ and every truncation tried. Cancellation between blocks never lowered the least weight. In the demand region that is $\nu_s/s \geq 2.08$, against the < 1 a joint lattice needs: a factor-two margin. This is evidence, not a proof: the true ν_s could be lower beyond $T = 13$, or for $s > 3$ (ν_s is subadditive, so small s does not bound the limit from below).

4. Run record

- `theorem`: `rustc -O --edition 2021 -D warnings examples/smallroot_optimal_reach.rs`, `smallroot_optimal_reach theorem > theorem_check.jsonl`, seed `0x7EE02026`, 5.6 s wall (`theorem_time.txt`).
- joint: `rustc … examples/smallroot_joint_lattice.rs`, `smallroot_joint_lattice > joint_nu.jsonl` (seed `0x4A012026`, 4.7 s, `joint_time.txt`) and `smallroot_joint_lattice --control > joint_control.jsonl`.
- Host: shared 4-core, 15 GB container; counts only, no timing claims.
- `scripts/checkpoint.sh push` cannot upload from this container (no S3 token).